

PROJECT III—TURBO OPTIONS FOR CRYPTOCURRENCIES

1 Objective

Price Turbo call and put options with Bitcoin / ETH as the underlying asset. Both European and American style options should be priced.

Delta exchange claimed to launch Turbo on Bitcoin/ETH. However, currently there are no Turbo options listed yet on Delta exchange. Turbo options specifics: <https://guides.delta.exchange/delta-exchange-user-guide/derivatives-guide/options-guide/turbo-options>

2 Data source

- Bitcoin / ETH price:
<https://finance.yahoo.com/quote/BTC-USD/history?p=BTC-USD>;
<https://finance.yahoo.com/quote/ETH-USD/history?p=ETH-USD>
- Options data: <https://www.deribit.com/options/BTC/BTC-12FEB26>

3 Questions

1. Download underlying asset's price data, analyze the data, estimate log return mean, volatility, and other parameter values. You may consider two risk-free rates $r = 1\%$ and $r = 5\%$ in your project.
2. Use standard binomial tree method (BTM) to Turbo call and put options in European style. Report the numerical results for different number of periods, strikes, Bitcoin/ETH prices, and time to maturities.
3. Use trinomial tree method (TTM) to price Asian call and put options in European style. Report the numerical results for different strikes, Bitcoin/ETH prices, and time to maturities.
4. Compare the convergence and computing efficiency of standard BTM and modified BTM.
5. Extend your pricing method and analysis to Turbo call and put options in American style.
6. Risk management analysis: compute the Greeks of Turbo options and compare the Greeks for Bitcoin and for ETH.
7. Note that there are only vanilla options (European options) on Bitcoin/ETH in the market. To persuade a platform (i.e., Deribit) to launch Turbo options, or other barrier options, and to persuade investors to trade them, please provide analysis on the pros and cons of Turbo options for Bitcoin/ETH. Justify your analysis with numerical results. For instance, you may use your pricing algorithm to provide comparison analysis between Turbo options and Vanilla options (for simplicity, you may use Black-Scholes formula to price vanilla options numerically).
8. Bitcoin / ETH prices have many jumps, will hedging with Turbo options help to mitigate such risk?

4 Requirements on project report

1. Introduction: a summary of research objective, methodology, and findings
2. Methodology and data: the pricing method, data description, and estimation method used in your project
3. Project findings: data analysis, pricing results and analysis, and discussions.
4. Conclusion.